

Notes: Melitz and Ottaviano (RES, 2008)

Market Size, Trade, and Productivity

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1 Big picture

Melitz and Ottaviano build a tractable trade model in which firm heterogeneity, endogenous mark-ups, and market-size effects all operate at the same time. The model keeps the selection logic of Melitz (2003), but replaces CES demand with linear demand. This change is the core move: it makes the toughness of competition endogenous. When a market becomes larger or more open, the number and productivity of competitors change, the maximum price a firm can charge falls, low-productivity firms are selected out, surviving firms lower mark-ups, and aggregate productivity rises.

The paper is about three related questions.

- How does market size affect the distribution of firm productivity, prices, mark-ups, output, revenue, and profits?
- How does costly trade differ from perfect integration? Does a larger country still have different firm outcomes after it trades with a smaller country?
- How do different liberalization policies - bilateral, unilateral, and preferential - affect competition, selection, entry location, mark-ups, and welfare?

The main answer is that market size and trade liberalization affect welfare through three linked channels: selection into production and export markets, pro-competitive mark-up reductions, and product-variety changes. Larger and more integrated markets are tougher markets: they have lower cost cut-offs, higher average productivity, lower prices, and lower mark-ups.

Economic message. The paper turns “trade raises productivity” into a precise market-competition mechanism. Trade does not only reallocate market shares toward more productive exporters. It also changes the residual demand elasticities faced by all firms, thereby changing mark-ups and the cut-off that determines which firms survive.

2 Incremental contribution of the paper

The paper’s incremental contribution is not that it discovers an entirely new source of gains from trade. Its contribution is to put several known channels into one highly tractable heterogeneous-firm model.

1. **Heterogeneous firms plus endogenous mark-ups.** Melitz (2003) has firm heterogeneity and selection, but CES demand fixes mark-ups. Melitz–Ottaviano keep firm heterogeneity but allow mark-ups to vary with competition.
2. **Market size matters even with trade.** In the closed economy, larger markets select tougher, more productive producers. In the open economy, costly trade does not fully integrate markets, so size differences continue to matter.
3. **A unified treatment of productivity, prices, mark-ups, and variety.** The model jointly predicts how market size and trade change average productivity, average mark-ups, firm size, price dispersion, product variety, and welfare.

4. **Policy experiments in a single framework.** Bilateral, unilateral, and preferential trade liberalization can all be studied with the same objects: domestic cost cut-offs, trade freeness, and entry patterns.
5. **A tractable multicountry extension.** The appendix shows that the model can be extended to many asymmetric countries with asymmetric trade costs and comparative advantage, while preserving closed-form expressions for cut-offs and a gravity-style trade equation.

Incremental contribution in one sentence. The paper transforms the Melitz selection mechanism into a model where competition itself is endogenous to market size and trade integration, so trade affects productivity both by selecting firms and by reducing mark-ups.

3 Data and measurement

This is a theory paper rather than an empirical paper. There is no empirical estimation, no original dataset, and no stand-alone table or figure in the source paper. The empirical role of the paper is indirect: it provides comparative statics that match several observed regularities documented in plant-level and establishment-level evidence, especially the patterns that larger markets have larger firms, higher average productivity, lower average prices, and less dispersion among low-productivity producers.

The model's key measurable objects are:

- **Firm marginal cost** c , which is the inverse of firm productivity.
- **Domestic cost cut-off** c_D , the marginal cost of the least productive firm that can profitably sell in a market.
- **Export cost cut-off** c_X , the marginal cost of the least productive firm that can profitably export.
- **Trade freeness** $\rho = \tau^{-k}$, where $\tau > 1$ is iceberg trade cost and k is the shape parameter of the productivity distribution.
- **Toughness of competition**, summarized by the price bound $p_{\max}$ and the cost cut-off c_D . A lower c_D means tougher selection and higher aggregate productivity.
- **Average productivity**, proxied by inverse average marginal cost, which increases when c_D falls.
- **Average mark-up**, which falls in larger or more integrated markets even though the least productive firms exit.

Measurement interpretation. In empirical applications, the model tells researchers to look for market-size and trade-cost effects not only on productivity but also on price dispersion, mark-up dispersion, entry/exit, and the distribution of firm size. The empirical prediction is distributional, not only aggregate.

4 Model environment

4.1 Preferences and linear demand

Consumers have quasi-linear preferences over a homogeneous numeraire and a continuum of differentiated varieties. The differentiated sector has quadratic utility, which implies linear demand. The key preference specification is

$$U = q_0^c + \alpha \int_{i \in \Omega} q_i^c di - \frac{1}{2} \gamma \int_{i \in \Omega} (q_i^c)^2 di - \frac{1}{2} \eta \left(\int_{i \in \Omega} q_i^c di \right)^2.$$

The parameter γ governs product differentiation. When γ is lower, varieties are closer substitutes. The parameter η governs substitutability between the differentiated sector and the numeraire. The inverse demand for each consumed variety is

$$p_i = \alpha - \gamma q_i^c - \eta Q^c.$$

The corresponding market demand is linear:

$$q_i = L q_i^c = \frac{\alpha L}{\eta N + \gamma} - \frac{L}{\gamma} p_i + \frac{\eta N}{\eta N + \gamma} \frac{L}{\gamma} \bar{p}.$$

Only varieties with prices below the choke price are consumed. The price bound is

$$p_{\max} = \frac{\gamma \alpha + \eta N \bar{p}}{\eta N + \gamma}.$$

Interpretation. A larger number of varieties N or a lower average price $\bar{p}$ reduces $p_{\max}$. A lower $p_{\max}$ means firms face more elastic residual demand. This is what the paper calls a tougher competitive environment.

4.2 Firm technology and pricing

Firms pay an irreversible entry cost f_E before learning their marginal cost c . Cost draws come from a distribution $G(c)$ on $[0, c_M]$. After observing c , a firm produces only if it can earn non-negative operating profit.

For a surviving firm, monopolistic competition implies

$$q(c) = \frac{L}{\gamma} [p(c) - c].$$

Let c_D be the marginal cost of the firm that just breaks even. This firm charges $p(c_D) = c_D = p_{\max}$ and sells zero. For any surviving firm $c < c_D$, firm outcomes are:

$$\begin{aligned}
p(c) &= \frac{1}{2}(c_D + c), & \mu(c) &= \frac{1}{2}(c_D - c), \\
q(c) &= \frac{L}{2\gamma}(c_D - c), & r(c) &= \frac{L}{4\gamma} \left[(c_D)^2 - c^2 \right], \\
\pi(c) &= \frac{L}{4\gamma}(c_D - c)^2.
\end{aligned}$$

Interpretation. More productive firms have lower marginal cost, lower prices, larger quantities, higher revenue, higher profits, and higher absolute mark-ups. The model therefore predicts that mark-ups vary within a market and are higher for more productive firms.

4.3 Free entry and the domestic cost cut-off

Free entry drives expected profit net of entry cost to zero:

$$\int_0^{c_D} \pi(c) dG(c) = \frac{L}{4\gamma} \int_0^{c_D} (c_D - c)^2 dG(c) = f_E.$$

The zero cut-off profit condition links the number of surviving firms to the cost cut-off:

$$N = \frac{2\gamma}{\eta} \frac{\alpha - c_D}{c_D - \bar{c}}.$$

A lower c_D means the least productive firms are excluded. Because average cost among survivors falls when c_D falls, a lower cut-off means higher aggregate productivity.

Main closed-economy comparative statics.

- Larger L lowers c_D , so larger markets have tougher selection.
- Lower entry cost f_E lowers c_D , because more entry strengthens competition.
- Lower product differentiation parameter γ lowers c_D , because varieties are closer substitutes and competition is stronger.
- Demand-shifters α and η affect the number of varieties but not the productivity cut-off under the Pareto parametrization.

4.4 Pareto productivity distribution

To obtain closed-form solutions, productivity $1/c$ follows a Pareto distribution. Equivalently,

$$G(c) = \left(\frac{c}{c_M} \right)^k, \quad c \in [0, c_M].$$

The shape parameter k governs dispersion. Higher k places more mass near high cost / low productivity draws.

The domestic cut-off becomes

$$c_D = \left[\frac{2(k+1)(k+2)\gamma(c_M)^k f_E}{L} \right]^{1/(k+2)}.$$

Average outcomes are simple functions of c_D . For example,

$$\begin{aligned} \bar{c} &= \frac{k}{k+1} c_D, & \bar{p} &= \frac{2k+1}{2k+2} c_D, \\ \bar{\mu} &= \frac{1}{2} \frac{1}{k+1} c_D, & \bar{\pi} &= f_E \frac{(c_M)^k}{(c_D)^k}. \end{aligned}$$

Welfare is

$$U = 1 + \frac{1}{2\eta}(\alpha - c_D) \left(\alpha - \frac{k+1}{k+2} c_D \right).$$

Economic message. Welfare is decreasing in the cost cut-off. Any policy or market feature that lowers c_D increases welfare through a mix of higher productivity, lower prices and mark-ups, and more variety.

4.5 Short-run versus long-run equilibrium

The paper distinguishes between long-run equilibrium, where entry and exit respond, and short-run equilibrium, where the number and distribution of incumbents are fixed. This distinction is crucial for asymmetric trade liberalization.

- In the **short run**, no new firms enter. Incumbent firms decide whether to produce or shut down. Changes in competition affect which incumbents operate, but they do not yet relocate entry across countries.
- In the **long run**, entry responds. A unilateral or preferential trade policy can shift where new firms choose to locate. These entry-location effects can overturn short-run gains.

Key distinction. The pro-competitive effect of import liberalization is a short-run benefit. The entry relocation effect is a long-run force that may benefit the non-liberalizing country and harm the liberalizing one.

5 Open-economy model

5.1 Two countries, segmented markets, and trade costs

There are two countries H and F , with L^H and L^F consumers. Markets are segmented. A firm produces in one country and sells domestically or abroad. To export to country l , a firm pays iceberg trade cost $\tau^l > 1$. The trade-cost index is

$$\rho^l = (\tau^l)^{-k} \in (0, 1),$$

where higher ρ means freer trade.

Let c_D^l be the cost cut-off for domestic producers in country l and c_X^l the export cut-off for firms from l to h . Because an exporter must cover the delivered cost of selling abroad,

$$c_X^h = \frac{c_D^l}{\tau^l}.$$

This means only relatively more productive firms export. A firm can survive domestically without being productive enough to export.

5.2 Open-economy free entry and domestic cut-offs

The two-country free-entry condition is

$$L^l (c_D^l)^{k+2} + L^h \rho^h (c_D^h)^{k+2} = \gamma \phi,$$

where

$$\phi = 2(k+1)(k+2)(c_M)^k f_E.$$

Solving the system gives the domestic cut-off in country l :

$$c_D^l = \left[\frac{\gamma \phi}{L^l} \frac{1 - \rho^h}{1 - \rho^l \rho^h} \right]^{1/(k+2)}.$$

Interpretation. A lower own trade barrier facing imports into country l raises import competition and lowers c_D^l in the short-run sense. In the full long-run solution, the domestic cut-off reflects own market size and bilateral integration, while firm entry reallocates across countries.

5.3 Prices, product variety, and welfare in the open economy

Domestic and foreign sellers in country l have matching price distributions once export costs are included. Thus average price and variety take the same functional forms as in the closed economy:

$$\bar{p}^l = \frac{2k+1}{2k+2} c_D^l, \quad N^l = \frac{2(k+1)\gamma}{\eta} \frac{\alpha - c_D^l}{c_D^l}.$$

Welfare is

$$U^l = 1 + \frac{1}{2\eta} (\alpha - c_D^l) \left(\alpha - \frac{k+1}{k+2} c_D^l \right).$$

Economic message. The domestic cut-off remains the sufficient statistic for welfare. If trade liberalization lowers c_D^l , welfare in country l rises.

5.4 Number of entrants and exporters

The number of entrants in country l is

$$N_E^l = \frac{2(c_M)^k(k+1)\gamma}{\eta(1-\rho^l\rho^h)} \left[\frac{\alpha - c_D^l}{(c_D^l)^{k+1}} - \rho^l \frac{\alpha - c_D^h}{(c_D^h)^{k+1}} \right].$$

Because entry responds to trade barriers, long-run effects are not simply the short-run effect of more import competition. Long-run entry can move across borders.

Selection into export markets. The appendix proves that in any non-specialized equilibrium with positive entry in both countries, $c_X^l < c_D^l$. Thus exporters are a strict subset of domestic producers and are more productive on average.

5.5 Reciprocal dumping

The model predicts reciprocal dumping even under monopolistic competition and differentiated products. Exporters set their free-on-board export price below their domestic price:

$$\frac{p_X^l(c)}{\tau^h} < p_D^l(c).$$

Economic message. Dumping here is not predatory pricing. It arises from market segmentation and different residual demand conditions across countries.

6 Trade liberalization experiments

6.1 Bilateral liberalization

With symmetric trade costs $\tau^H = \tau^F = \tau$ and $\rho^H = \rho^F = \rho$, the domestic cut-off becomes

$$c_D^l = \left[\frac{\gamma\phi}{L^l(1+\rho)} \right]^{1/(k+2)}.$$

A bilateral increase in trade freeness ρ lowers cost cut-offs in both countries. Thus bilateral liberalization:

- raises aggregate productivity in both countries;
- forces the least productive firms out;
- reallocates market shares toward more productive firms;
- lowers average mark-ups and prices;

- raises product variety;
- increases welfare in both countries.

Market-size asymmetry. If $L^l > L^h$, then country l has a lower cut-off and more firms. Bilateral liberalization reduces the cut-off differential but increases the cross-country differences in number of sellers, entrants, and producers. The larger market becomes even more attractive as an entry location.

6.2 Unilateral liberalization

A unilateral liberalization by country l means ρ^l rises while ρ^h is fixed. The paper emphasizes the distinction between short-run and long-run effects.

Short run. With the number of incumbents fixed, unilateral liberalization increases import competition in country l . The cost cut-off in l falls, less productive incumbents shut down, product variety increases, mark-ups fall, and welfare in l rises. The trading partner is unaffected in the short run.

Long run. Entry shifts toward the non-liberalizing country. In the long-run free-entry equilibrium, the liberalizing country can end up with weaker competition and a higher cost cut-off, while the partner has stronger competition and a lower cut-off. Thus the short-run gains from unilateral liberalization can be overturned in the long run.

Economic message. The pro-competitive effect and the entry-location effect push in different directions. A country that unilaterally opens its market gains from import competition initially, but may lose if entry relocates abroad.

6.3 Preferential liberalization

The paper extends the model to three countries H , F , and T . A preferential trade agreement between H and F raises their mutual trade freeness but leaves trade costs with T unchanged.

The three-country cut-off equation is

$$(c_D^l)^{k+2} + \rho^{lh}(c_D^h)^{k+2} + \rho^{ht}(c_D^t)^{k+2} = \frac{\gamma\phi}{L}.$$

With initially symmetric barriers, the cut-offs are

$$\begin{aligned} c_D &= \left[\frac{\gamma\phi}{L} \frac{1}{1+2\rho} \right]^{1/(k+2)}, \\ c_D^H &= c_D^F = \left[\frac{\gamma\phi}{L} \frac{1-\rho}{1-2\rho^2+\rho'} \right]^{1/(k+2)}, \\ c_D^T &= \left[\frac{\gamma\phi}{L} \frac{(1-\rho) + (\rho' - \rho)}{1-2\rho^2+\rho'} \right]^{1/(k+2)}. \end{aligned}$$

Preferential liberalization lowers cut-offs in the liberalizing countries H and F and raises the cut-off in the excluded country T .

Economic message. Preferential agreements make participating countries better export bases. The excluded country can lose because entry and competition shift toward the integrated bloc.

7 Appendix logic and extensions

The appendix is important because it shows that the model is not limited to a stylized two-country world.

7.1 Variance of firm performance measures

Given the Pareto cost distribution, the variance of prices, quantities, mark-ups, and revenues can all be written as simple functions of the variance of cost draws:

$$\sigma_p^2 = \frac{1}{4}\sigma_c^2, \quad \sigma_q^2 = \frac{L^2}{4\gamma^2}\sigma_c^2, \quad \sigma_\mu^2 = \frac{1}{4}\sigma_c^2, \quad \sigma_r^2 = \frac{L^2}{16\gamma^2}\sigma_c^2.$$

Interpretation. Larger markets reduce the support of the cost distribution through selection, so they reduce dispersion in cost, prices, and mark-ups. They may increase dispersion in output and revenue because market size magnifies differences among survivors.

7.2 Multiple countries and comparative advantage

With M countries and asymmetric trade costs, the domestic cut-off is

$$c_D^l = \left[\frac{2(k+1)(k+2)f_E\gamma}{|P|} \frac{\sum_{h=1}^M |C^{hl}|/\psi^h}{L^l} \right]^{1/(k+2)},$$

where P is the trade-freeness matrix, $|C^{hl}|$ is the cofactor of element ρ^{hl} , and $\psi^h = (c_M^h)^{-k}$ captures comparative advantage.

This expression shows that cross-country cut-off differences come from three sources: own market size, market access, and comparative advantage.

7.3 Gravity equation

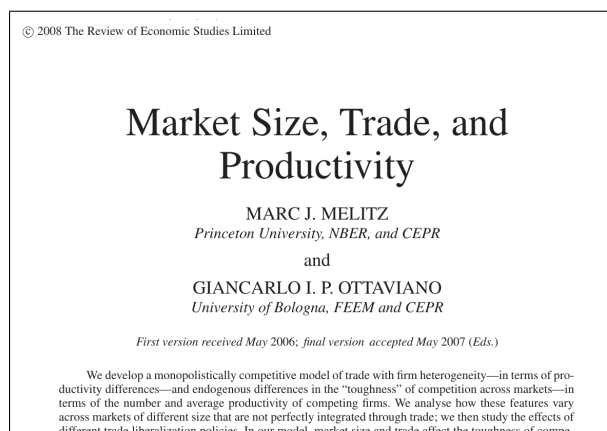
The appendix also derives a gravity-style equation for bilateral exports:

$$EXP^{lh} = \frac{1}{2\gamma(k+2)} N_E^l \psi^l L^h (c_D^h)^{k+2} (\tau^{lh})^{-k}.$$

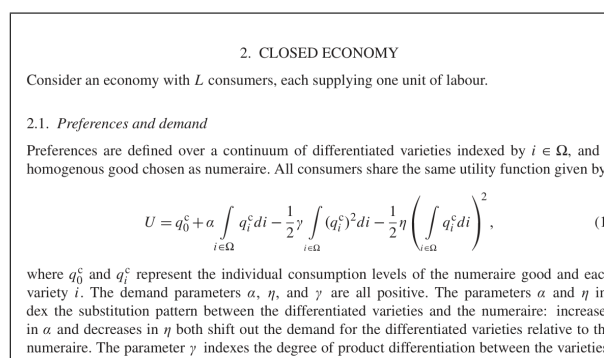
Interpretation. Bilateral exports depend on exporter entry, exporter technology, importer market size, importer competition, and bilateral trade barriers. Tougher competition in the importing country makes it harder for exporters to enter, dampening trade flows.

8 Tables and figures from the paper

The source paper contains no stand-alone numbered tables or figures. To follow the same visual-note standard, I directly extracted the main model-equation panels from the PDF. Each pair below is placed on the same row. The panels are treated as visual excerpts of the paper’s core theoretical objects.



Extracted panel: title and abstract.



Extracted panel: utility function and closed-economy setup.

Read. The abstract states the paper’s central mechanism: market size and trade affect competition, competition affects selection, and selection changes productivity and mark-ups.

Economic message. The utility block is the foundation for endogenous mark-ups. Linear demand is what makes competition tougher when N rises or average prices fall.

increases with γ as consumers give increasing weight to the distribution of consumption level across varieties. The marginal utilities for all goods are bounded, and a consumer may thus not have positive demand for any particular good. We assume that consumers have positive demands for the numeraire good ($q_0^c > 0$). The inverse demand for each variety i is then given by $p_i = \alpha - \gamma q_i^c - \eta Q^c, \quad (2)$ whenever $q_i^c > 0$. Let $\Omega^* \subset \Omega$ be the subset of varieties that are consumed ($q_i^c > 0$). Equation (2) can then be inverted to yield the linear market demand system for these varieties: $q_i \equiv L q_i^c = \frac{\alpha L}{\eta N + \gamma} - \frac{L}{\gamma} p_i + \frac{\eta N}{\eta N + \gamma} \frac{L}{\gamma} \bar{p}, \quad \forall i \in \Omega^*, \quad (3)$ where N is the measure of consumed varieties in Ω^* and $\bar{p} = (1/N) \int_{i \in \Omega^*} p_i di$ is their average price. The set Ω^* is the largest subset of Ω that satisfies $p_i \leq \frac{1}{\eta N + \gamma} (\gamma \alpha + \eta N \bar{p}) \equiv p_{\max}, \quad (4)$ where the R.H.S. price bound $p_{\max}$ represents the price at which demand for a variety is driven to 0. Note that (2) implies $p_{\max} \leq \alpha$. In contrast to the case of Constant Elasticity of Substitution (CES) demand, the price elasticity of demand, $\varepsilon_i \equiv (\partial q_i / \partial p_i)(p_i / q_i) = [(p_{\max} / p_i) - 1]^{-1}$, is not uniquely determined by the level of product differentiation γ. Given the latter, lower average prices $\bar{p}$ or a larger number of competing varieties N induce a decrease in the price bound $p_{\max}$ and an increase in the price elasticity of demand ε_i at any given p_i. We characterize this as “tougher” competitive environment.³ Welfare can be evaluated using the indirect utility function associated with (1):	
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Extracted panel: linear demand, choke price, and welfare.

satisfy $q(c) = \frac{L}{\gamma} [p(c) - c]. \quad (6)$ The profit maximizing price $p(c)$ may be above the price bound $p_{\max}$ from (4), in which case the firm exits. Let c_D reference the cost of the firm who is just indifferent about remaining in the industry. This firm earns zero profit as its price is driven down to its marginal cost, $p(c_D) = c_D = p_{\max}$, and its demand level $q(c_D)$ is driven to 0. We assume that c_M is high enough to be above c_D, so that some firms with cost draws between these two levels exit. All firms with cost $c < c_D$ earn positive profits (gross of the entry cost) and remain in the industry. The threshold cost c_D summarizes the effects of both the average price and number of firms on the performance measures of all firms. Let $r(c) = p(c)q(c)$, $\pi(c) = r(c) - q(c)c$, $\mu(c) = p(c) - c$ denote the revenue, profit, and (absolute) mark-up of a firm with cost c. All these performance measures can then be written as functions of c and c_D only: $p(c) = \frac{1}{2}(c_D + c), \quad (7)$ $\mu(c) = \frac{1}{2}(c_D - c), \quad (8)$ $q(c) = \frac{L}{2\gamma}(c_D - c), \quad (9)$ $r(c) = \frac{L}{4\gamma}[(c_D)^2 - c^2], \quad (10)$ $\pi(c) = \frac{L}{4\gamma}(c_D - c)^2. \quad (11)$ As expected, lower cost firms set lower prices and earn higher revenues and profits than firms with higher costs. However, lower cost firms do not pass on all of the cost differential to consumers in the form of lower prices: they also set higher mark-ups (in both absolute and relative terms) than firms with higher costs.	
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Extracted panel: firm pricing, quantities, revenues, profits, and mark-ups.

Read. Demand depends on N and $\bar{p}$, and firm outcomes depend on the cut-off c_D .

Economic message. The model’s sufficiency result begins here: once c_D is known, we can characterize prices, quantities, mark-ups, revenue, profit, and welfare.

which determines the cost cut-off c_D. This cut-off, in turn, determines the number of surviving firms, since $c_D = p(c_D)$ must also be equal to the zero demand price threshold in (4): $c_D = \frac{1}{\eta N + \gamma} (\gamma \alpha + \eta N \bar{p}).$ This yields the zero cut-off profit condition $N = \frac{2\gamma}{\eta} \frac{\alpha - c_D}{c_D - \bar{c}}, \quad (13)$ where $\bar{c} = [\int_0^{c_D} c dG(c)] / G(c_D)$ is the average cost of surviving firms.⁵ The number of entrant is then given by $N_E = N / G(c_D)$. Given a production technology referenced by $G(c)$, average productivity will be higher (lower $\bar{c}$) when sunk costs are lower, when varieties are closer substitutes (lower γ), and in bigger markets (more consumers L). In all these cases, firm exit rates are also higher (the pre entry probability of survival $G(c_D)$ is lower). The demand parameters α and η that index the overall level of demand for the differentiated varieties (relative to the numeraire) do not affect the selection of firms and industry productivity—they only affect the equilibrium number of firms. Competition is “tougher” in larger markets as more firms compete and average price $\bar{p} = (c_D + \bar{c})/2$ are lower. A firm with cost c responds to this tougher competition by setting lower mark-up (relative to the mark-up it would set in a smaller market—see (8)). 2.4. Parametrization of technology All the results derived so far hold for any distribution of cost draws $G(c)$. However, in order to simplify some of the ensuing analysis, we use a specific parametrization for this distribution. In particular, we assume that productivity draws $1/c$ follow a Pareto distribution with low productivity bound $1/c_M$ and shape parameter $k \geq 1$.⁶ This implies a distribution of cost draws given by $G(c) = \left(\frac{c}{c_M} \right)^k, \quad c \in [0, c_M]. \quad (14)$ The shape parameter k indexes the dispersion of cost draws. When $k = 1$, the cost distribution is uniform on $[0, c_M]$. As k increases, the relative number of high-cost firms increases	
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Extracted panel: free entry, zero cut-off profit, and Pareto technology.

previously anticipated. The number of surviving firms, determined by (13), is then $N = \frac{2(k+1)\gamma}{\eta} \frac{\alpha - c_D}{c_D}. \quad (16)$ This parametrization also yields simple derivations for the averages of all the firm-level performance measures described in (7)–(11):⁷ $\bar{c} = \frac{k}{k+1} c_D, \quad \bar{q} = \frac{L}{2\gamma} \frac{1}{k+1} c_D = \frac{(k+2)(c_M)^k}{(c_D)^{k+1}} f_E,$ $\bar{p} = \frac{2k+1}{2k+2} c_D, \quad \bar{r} = \frac{L}{2\gamma} \frac{1}{k+2} (c_D)^2 = \frac{(k+1)(c_M)^k}{(c_D)^k} f_E,$ $\bar{\mu} = \frac{1}{2} \frac{1}{k+1} c_D, \quad \bar{\pi} = f_E \frac{(c_M)^k}{(c_D)^k}.$ As with the cost average $\bar{c}$, the average for a performance measure $z(c)$ is given by $\bar{z} = [\int_0^{c_D} z(c) dG(c)] / G(c_D)$. Although $\bar{c}$ is computed as the unweighted average of firm cost, it provides an index to a much broader set of inverse productivity measures. The average of firm productivity $1/c$—whether unweighted, weighted by revenue $r(c)$, or weighted by output $q(c)$—is proportional to $1/\bar{c}$ (and hence to $1/c_D$). In the Appendix, we further show how the variance of all the firm performance measures can be written as simple functions of the variance of the cost draws. Since the cut-off level completely summarizes the distribution of prices as well as all the other performance measures, it also uniquely determines welfare from (5): $U = 1 + \frac{1}{2\eta} (\alpha - c_D) \left(\alpha - \frac{k+1}{k+2} c_D \right). \quad (17)$ Welfare increases with decreases in the cut-off c_D, as the latter induces increases in produc	
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Extracted panel: closed-form averages and welfare under Pareto costs.

Read. Free entry determines c_D ; the Pareto assumption produces tractable averages.

Economic message. Larger markets lower c_D . Lower c_D simultaneously means higher average productivity, lower average prices, lower average mark-ups, and higher welfare.

cost cut-off rules for firms selling in either market. Let c_D^l denote the upper bound cost for firm selling in their domestic market, and let c_X^l denote the upper bound cost for exporters from l to h . These cut-offs must then satisfy

$$\begin{aligned} c_D^l &= \sup\{c : \pi_D^l(c) > 0\} = p_{\max}^l, \\ c_X^l &= \sup\{c : \pi_X^l(c) > 0\} = \frac{p_{\max}^h}{\tau^h}. \end{aligned} \quad (19)$$

This implies $c_X^h = c_D^l / \tau^l$: trade barriers make it harder for exporters to break even relative to domestic producers.

As was the case in the closed economy, the cut-offs summarize all the effects of market conditions relevant for firm performance. In particular, the optimal prices and output levels can be written as functions of the cut-offs:

$$\begin{aligned} p_D^l(c) &= \frac{1}{2}(c_D^l + c), \quad q_D^l(c) = \frac{L^l}{2\gamma}(c_D^l - c), \\ p_X^l(c) &= \frac{\tau^l}{2}(c_X^l + c), \quad q_X^l(c) = \frac{L^h}{2\gamma}\tau^h(c_X^l - c), \end{aligned} \quad (20)$$

which yield the following maximized profit levels:

$$\begin{aligned} \pi_D^l(c) &= \frac{L^l}{4\gamma}(c_D^l - c)^2, \\ \pi_X^l(c) &= \frac{L^h}{4\gamma}(\tau^h)^2(c_X^l - c)^2. \end{aligned} \quad (21)$$

3.1. Free entry condition

Entry is unrestricted in both countries. Firms choose a production location prior to entry and paying the sunk entry cost. In order to focus our analysis on the effects of market size and trade costs differences, we assume that countries share the same technology—referenced by the entry cost f_E and cost distribution $G(c)$.¹² Free entry of domestic firms in country l implies zero expected profits in equilibrium, hence

Extracted panel: open-economy cut-offs and export-profit expressions.

(22) can be re-written

$$L^l (c_D^l)^{k+2} + L^h \rho^h (c_D^h)^{k+2} = \gamma \phi,$$

where $\rho^l \equiv (\tau^l)^{-k} \in (0, 1)$ is an inverse measure of trade costs (the “freeness” of trade). This system (for $l = H, F$) can then be solved for the cut-offs in both countries:

$$c_D^l = \left[\frac{\gamma \phi}{L^l} \frac{1 - \rho^h}{1 - \rho^l \rho^h} \right]^{1/(k+2)}. \quad (23)$$

3.2. Prices, product variety, and welfare

The prices in country l reflect both the domestic prices of country- l firms, $p_D^l(c)$, and the price of exporters from h , $p_X^h(c)$. Using (19) and (20), these prices can be written:

$$\begin{aligned} p_D^l(c) &= \frac{1}{2}(p_{\max}^l + c), \quad c \in [0, c_D^l], \\ p_X^h(c) &= \frac{1}{2}(p_{\max}^h + \tau^h c), \quad c \in [0, c_D^l / \tau^h], \end{aligned}$$

where $p_{\max}^l$ is the price threshold defined in (18). In addition, the cost of domestic firms $c \in [0, c_D^l]$ and the delivered cost of exporters $\tau^h c \in [0, c_D^h]$ have identical distributions over this support, given by $G^l(c) = (c/c_D^l)^k$. The price distribution in country l of domestic firms producing in l , $p_D^l(c)$, and exporters producing in h , $p_X^h(c)$, are therefore also identical. The average price in country l is thus given by

$$\bar{p}^l = \frac{2k+1}{2k+2} c_D^l.$$

Combining this with the threshold price in (18) determines the number of firms selling in

Extracted panel: open-economy domestic cut-off and welfare.

Read. Trade costs create separate domestic and export cut-offs; welfare still depends on the domestic cut-off.

Economic message. The open economy preserves the tractability of the closed economy. Trade affects welfare through the same core object, c_D^l .

$G(c_D^l)N_E^l + G(c_X^h)N_E^h = N^l$. This condition (holding for each country) can be solved for the number of entrants in each country:

$$\begin{aligned} N_E^l &= \frac{(c_M)^k}{1 - \rho^l \rho^h} \left[\frac{N^l}{(c_D^l)^k} - \rho^l \frac{N^h}{(c_D^h)^k} \right] \\ &= \frac{2(c_M)^k(k+1)\gamma}{\eta(1 - \rho^l \rho^h)} \left[\frac{\alpha - c_D^l}{(c_D^l)^{k+1}} - \rho^l \frac{\alpha - c_D^h}{(c_D^h)^{k+1}} \right]. \end{aligned} \quad (26)$$

In the Appendix, we show that (26) further implies that $c_X^l < c_D^l$ in this non-specialized equilibrium ($N_E^l > 0$), so that only a subset of relatively more productive firms export. The remaining higher cost firms (with cost between c_X^l and c_D^l) only serve their domestic market. $G(c_D^l)N_E^l$ thus also represents the total number of firms producing in l (no firm produces in l without also serving its domestic market).

3.4. Reciprocal dumping and arbitrage opportunities

Brander and Krugman (1983) have shown that reciprocal dumping must occur in intra-industry trade equilibria under Cournot competition where representative firms produce a single homogeneous good in two countries. Ottaviano *et al.* (2002) show that dumping can also occur with differentiated products under monopolistic competition with representative firms. We extend this result to our current framework, where firms with heterogeneous costs produce differentiated varieties and face different residual demand price elasticities.¹⁵ Given the optimal price functions $p_D^l(c) = (c_D^l + c)/2$ and $p_X^l(c) = \tau^h(c_X^l + c)/2$ from (20), $c_X^l < c_D^l$ implies that

cost cut-off c_D . When analysing the impact of trade, we can therefore focus on the determination of the cost cut-off governed by (23).

Comparing this new cut-off condition to the one derived for the closed economy (15) immediately reveals that the cost cut-off is lower in the open economy: trade increases aggregate productivity by forcing the least productive firms to exit. This effect is similar to that analysed in Melitz (2003) but works through a different economic channel. In Melitz (2003), trade induces increased competition for scarce labour resources as real wages are bid up by the relatively more productive firms who expand production to serve the export markets. The increase in real wage forces the least productive firms to exit. In that model, import competition does not play a role in the reallocation process due to the CES specification for demand (residual demand price elasticities are exogenously fixed and unaffected by import competition). In the current model, the impact of these two channels—via increased factor market or product market competition—is reversed: increased product market competition is the only operative channel. Increased factor market competition plays no role in the current model, as the supply of labour to the differentiated goods sector is perfectly elastic. On the other hand, import competition increases competition in the domestic product market, shifting up residual demand price elasticities for all firms at any given demand level. This forces the least productive firms to exit. This effect is very similar to an increase in market size in the closed economy: the increased competition induces a downward shift in the distribution of mark-ups across firms. Although only relatively more productive firms survive (with higher mark-ups than the less productive firms who exit), the average mark-up is reduced. The distribution of prices shifts down due to the combined effect of selection and lower mark-ups. Again, as in the case of larger market size in a closed economy, average firm size and profits increase—as does product variety.¹⁷ In this model, welfare gains from trade thus come from a combination of productivity gains (via selection), lower mark-ups (pro-competitive effect), and increased product variety.

3.6. Market size effects

We now focus on the consequences of market size differences for cross-country characteristics in the open-economy equilibrium. Once again, these cross-country differences in firm performance measures will be determined by the differences in the cost cut-offs c_D^l , as shown in (23). This immediately highlights how costly trade does not completely integrate markets as respective country size plays an important role in determining all firm performance measures and welfare.

Extracted panel: number of entrants and reciprocal dumping.

Extracted panel: trade impact and market-size effects.

Read. Entry responds to relative market conditions; exporters are selected from the more productive part of the cost distribution.

Economic message. The model separates the productivity gains from selection from the pro-competitive gains from lower mark-ups. Both operate in the same market equilibrium.

This location decision also crucially affects the long-run consequences of trade liberalization—especially in situations where the decreases in trade barriers are asymmetric.

4.1. Bilateral liberalization

Before illustrating the consequences of asymmetric liberalization, we first quickly describe the case of symmetric liberalization. Here, we assume that trade costs are symmetric, $\tau^H = \tau^F = \tau$ and analyse the effects of decreases in τ (increases in $\rho^H = \rho^F = \rho$). In this case, the equilibrium cut-off condition (23) can be written:

$$c_D^l = \left[\frac{\gamma \phi}{L^l(1+\rho)} \right]^{1/(k+2)}. \quad (29)$$

Bilateral liberalization thus increases competition in both markets, leading to proportional changes in the cut-offs (and hence proportional increases in aggregate productivity) in both countries.²¹ The effects of such liberalization are thus qualitatively identical to those described for the transition from autarky to the open economy: Product variety increases as a result of the increased competition, which also induces a decrease in mark-ups and prices. Again, welfare rises from a combination of higher productivity, lower mark-ups, and increased product variety.

Symmetric trade liberalization induces all of the same qualitative results in the short run. Although these effects do not depend on relative country size (so long as the differentiated good is produced in both countries), differences in country size nevertheless induce important changes in the relative pattern of entry in the long run following liberalization. Assuming $L^l > L^h$, the positive entry differential $N_E^l - N_E^h$ widens with liberalization as entry in the bigger market becomes relatively more attractive. This also induces a growing differential in the number of domestic producers $N_D^l - N_D^h$ (see Appendix for proofs).

4.2. Unilateral liberalization

We now describe the effects of a unilateral liberalization by country l (an increase in ρ^l , holding ρ^h constant). Given the cut-off condition (23), this leads to an increase in the cost cut-off c_D^l (less competition in the liberalizing country)—whereas the cut-off c_D^h in the country's trading partner decreases, indicating an increase in competition there. The liberalizing country thus experiences

the lowest sum of bilateral barriers (highest $\rho^{lh} + \rho^{ll}$). In effect, this country is the best export base or “hub”. Moreover, since ρ^{ht} enters the expression of the cut-off for country l , any change in bilateral trade costs affects all three countries. This has important implications for preferential trade agreements.²⁴ To see this as clearly as possible, consider three countries with initial symmetric trade barriers ($\rho^{lt} = \rho$). The initial cut-offs are then identical and equal to (see (30))

$$c_D = \left[\frac{\gamma \phi}{L} \frac{1}{1+2\rho} \right]^{1/(k+2)}. \quad (31)$$

A preferential trade agreement is then introduced between H and F , inducing $\rho^{HF} = \rho' > \rho = \rho^{FT} = \rho^{HT}$. The new trade regime affects the cut-offs for all countries. From (30), the cut-offs in the liberalizing countries are then

$$c_D^H = c_D^F = \left[\frac{\gamma \phi}{L} \frac{1-\rho}{1-2\rho^2+\rho'} \right]^{1/(k+2)}, \quad (32)$$

while the cut-off in the third country is given by

$$c_D^T = \left[\frac{\gamma \phi}{L} \frac{(1-\rho) + (\rho' - \rho)}{1-2\rho^2+\rho'} \right]^{1/(k+2)}. \quad (33)$$

The number of entrants in all three countries are given by (recall that $c_D^H = c_D^F$)

$$N_E^H = N_E^F = \frac{2(c_M)^k(k+1)\gamma}{\eta(1-2\rho^2+\rho')} \left[\frac{\alpha - c_D^H}{(c_D^H)^{k+1}} - \rho \frac{\alpha - c_D^T}{(c_D^T)^{k+1}} \right],$$

$$N_E^T = \frac{2(c_M)^k(k+1)\gamma}{\eta(1-2\rho^2+\rho')} \left[(1+\rho') \frac{\alpha - c_D^T}{(c_D^T)^{k+1}} - 2\rho \frac{\alpha - c_D^H}{(c_D^H)^{k+1}} \right].$$

Comparing (31)–(33), it is easily verified that preferential liberalization leads to lower cut-offs in the liberalizing countries and a higher cut-off in the third country. Thus, average costs, prices, and mark-ups also decrease in the liberalizing countries while they rise in the third country. The liberalizing countries become better “export bases”: they gain better access to each other's market while maintaining the same ease of access to the third country's market. Thus, preferential liberalization leads to long-run welfare gains for the liberalizing countries alone.

Extracted panel: bilateral and unilateral liberalization.

Extracted panel: preferential liberalization in a three-country world.

Read. Bilateral liberalization lowers cut-offs in both countries. Unilateral and preferential liberalization require care because entry relocates in the long run.

Economic message. Symmetric liberalization is unambiguously pro-competitive. Asymmetric liberalization can generate short-run gains but long-run losses through firm relocation.

tion resulting in lower average mark-ups and higher aggregate productivity. We also show how costly trade does not completely integrate markets and thus does not obviate these important consequences of market size differences across trading partners.

We then analyse several different trade liberalization scenarios. Our model highlights the pro-competitive effects of increased import competition and its effect on mark-ups, productivity and product variety in the liberalized import market. Our model also echoes the findings in previous work that show how the short-run gains of asymmetric liberalization can be reversed by shifts in the pattern of entry in the long run. However, our model additionally incorporates the important feedbacks between entry and firm selection into domestic and export markets.

Although each of these individual channels for trade-induced gains have been previously analysed in models with different structures, we believe it is important to show how all of these channels can be captured within a single unified framework. This framework develops a new and very tractable way of describing how differences in market size and trade costs across trading partners affect the distribution of key firm-level performance measures across markets. We hope that this provides a useful foundation for future empirical investigations.

APPENDIX

A.1. Variance of firm performance measures

Let $\sigma_c^2 = \left[\int_0^D (c - \bar{c})^2 dG(c) \right] / G(c_D)$ denote the variance of the firm cost draws. As was mentioned earlier in the main text, the variance of all the firm performance measures can be written as simple expressions of this variance:

$$\begin{aligned} \sigma_p^2 &= \frac{1}{4} \sigma_c^2, \quad \sigma_q^2 = \frac{L^2}{4\gamma^2} \sigma_c^2, \\ \sigma_\mu^2 &= \frac{1}{4} \sigma_c^2, \quad \sigma_r^2 = \frac{L^2}{16\gamma^2} \sigma_c^2, \end{aligned}$$

where $\sigma_z^2 = \left[\int_0^D [z(c) - \bar{z}]^2 dG(c) \right] / G(c_D)$ and $\bar{z}$ denote the variance and mean of a firm performance measure $z(c)$.

draws. For tractability, we assume that firm productivity $1/c$ is distributed Pareto with shape k in all countries, but allow for differences in the support of the distributions via differences in the upper-bound cost c_M^l . The cost draws in country l thus have a distribution $G^l(c) = (c/c_M^l)^k$. Whenever $c_M^l < c_M^h$, country l will have a comparative advantage with respect to country h in the differentiated good sector: entrants in country l have a better chance of getting higher productivity draws.²⁵

In this extended model, the free entry condition (22) in country l becomes

$$\sum_{h=1}^M \rho^{lh} L^h (c_D^h)^{k+2} = \frac{2\gamma(k+1)(k+2)f_E}{\psi^l}, \quad l = 1, \dots, M,$$

where $\psi^l = (c_M^l)^{-k}$ is an index of comparative advantage. This yields a system of M equations that can be solved for the M equilibrium domestic cut-offs using Cramer's rule:

$$c_D^l = \left(\frac{2(k+1)(k+2)f_E\gamma}{|P|} \sum_{h=1}^M |C_{hl}| / \psi^h \right)^{1/(k+2)}, \quad (\text{A.1})$$

where $|P|$ is the determinant of the trade freeness matrix

$$P = \begin{pmatrix} \rho_{11} & \rho_{12} & \dots & \rho_{1M} \\ \rho_{21} & \rho_{22} & \dots & \rho_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \rho_{M1} & \rho_{M2} & \dots & \rho_{MM} \end{pmatrix},$$

and $|C_{hl}|$ is the cofactor of its ρ_{hl} element. Cross-country differences in cut-offs now arise from three sources: own country size (L^l), as well as a combination of market access and comparative advantage ($\sum_{h=1}^M |C_{hl}| / \psi^h$). Countries benefiting from a larger local market, a better distribution of productivity draws, and better market accessibility have lower cut-offs.

The mass of sellers N^l in each country l (including domestic producers in l and exporters to l) is still given by (24). Given a positive mass of entrants N_E^l in all countries, there are $G^l(c_D^l)N_E^l$ domestic producers and $\sum_{h \neq l} G^l(c_X^h)N_l$ exporters selling in l , where c_X^h is the export cut-off from h to l . This implies

$$\sum_{h=1}^M \rho^{hl} \psi^h N_E^h = \frac{N^l}{(c_D^l)^k}.$$

The latter provides a system of M linear equations that can be solved for the number of entrants in the M countries using Cramer's rule:²⁶

$$N_E^l = \frac{2(k+1)\gamma}{\eta |P| \psi^l} \sum_{h=1}^M \frac{(\alpha - c_D^h) |C_{hl}|}{(c_D^h)^{k+1}}. \quad (\text{A.2})$$

Given N_E^l entrants in country l , $N_E^l G^l(c_D^l)$ firms survive and produce for the local market. Among the latter $N_E^l G^l(c_X^h)$ export to country h .

Extracted panel: conclusion and appendix variance formulas.

Extracted panel: multicountry extension and gravity equation.

Read. The conclusion stresses that a single framework captures productivity, mark-up, price, and size responses. The appendix generalizes the model to asymmetric multicountry trade.

Economic message. The paper is designed as a reusable modelling toolkit. The main economics survives beyond the two-country case.

9 Short summary

Melitz and Ottaviano develop a heterogeneous-firm trade model with endogenous mark-ups. The key modelling move is to use a linear demand system, which makes the residual demand elasticity faced by firms depend on market conditions. Larger markets and more integrated markets are tougher: they have more competitors and lower average prices, which lowers the maximum price a producer can charge. As a result, the least productive firms exit, average productivity rises, and surviving firms charge lower average mark-ups.

In the closed economy, larger markets select more productive firms, have more product variety, lower average prices and mark-ups, and higher welfare. In the open economy, costly trade does not fully integrate markets, so country size still matters. The larger country continues to have a lower productivity cut-off, more productive firms, lower prices, and lower mark-ups.

For trade policy, bilateral liberalization has the cleanest implication: it increases competition in both countries and raises welfare. Unilateral liberalization is more subtle. In the short run it benefits the liberalizing country through more import competition, but in the long run entry may

relocate to the non-liberalizing partner and overturn the initial gain. Preferential liberalization similarly benefits the participating countries in the long run but may hurt the excluded country by making it a worse export base.

10 Conclusion

10.1 Core conclusion

The paper's core conclusion is that market size and trade integration affect aggregate productivity through the toughness of competition. Larger and more open markets force out low-productivity firms and reduce average mark-ups. These forces make productivity, prices, mark-ups, variety, and welfare endogenous to market size and trade costs.

10.2 Economic intuition

The intuition is simple. When a market becomes larger or more exposed to imports, firms face tougher residual demand. The price bound $p_{\max}$ falls. Firms with high marginal costs can no longer profitably sell. Surviving firms are more productive on average, but they also face stronger competition, so average mark-ups fall. The gains from trade therefore come from both selection and pro-competitive effects.

10.3 Policy interpretation

The model implies that trade liberalization is not one-dimensional. Bilateral liberalization is broadly beneficial because both markets become tougher. Unilateral liberalization and preferential liberalization must be evaluated dynamically because entry responds. In the short run, import competition raises welfare in the liberalizing market. In the long run, entry relocation can shift the gains toward trade partners or toward members of a preferential bloc.

10.4 Limitations

The model gains tractability from strong assumptions: quasi-linear preferences, linear demand, Pareto productivity, no fixed export costs, constant marginal production costs, and a focus on the differentiated sector with a numeraire. These assumptions are useful for closed-form results, but they also shape the policy conclusions. In particular, the absence of fixed export costs makes export selection driven by variable trade costs, and the quasi-linear structure avoids income effects.

10.5 Incremental contribution revisited

The incremental contribution is that the paper unifies Melitz-style selection with endogenous mark-ups and market-size-dependent competition. It turns the trade-productivity connection into a tractable framework that can analyze not only bilateral liberalization but also unilateral and preferential policy changes. This makes the paper a central bridge between heterogeneous-firm trade theory and the older pro-competitive-effects literature.

10.6 Takeaway

The most important takeaway is: **trade changes productivity because it changes the toughness of competition.** In this model, larger or more integrated markets are tougher markets; tougher markets select better firms and compress mark-ups; and the domestic cost cut-off c_D is the central object summarizing these effects.